# The stationary-start maintenance model and its exact delay-period predictions

Working note

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**Abstract**

One change to the maintenance model — initialising the Ornstein–Uhlenbeck maintenance noise at its stationary law rather than at zero — makes the transient irrelevant to the delay-period analysis window, and every prediction the paper tests then follows from the *stationary* Kac–Rice rate by a short corollary. The note is organised accordingly: Sections 1–4 establish that the transient can be neglected, and state precisely under what conditions; Sections 5–7 work with the stationary rate alone and derive the predictions. Every statement has been verified numerically (Section 9).

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## 1 The model

Everything before stimulus offset is unchanged. Maintenance begins at $\tau = t_{\text{offset}} + L_{\text{motion}}$, at which time the gate-facing signal $y$ equals the maintained target $m$. During maintenance,

$$\mathrm{d}\varepsilon(t) = -\kappa\,\varepsilon(t)\,\mathrm{d}t + \sigma_\varepsilon\,\mathrm{d}W_\varepsilon(t), \tag{1}$$

$$\mathrm{d}y(t) = \lambda\,[\,m + \varepsilon(t) - y(t)\,]\,\mathrm{d}t, \tag{2}$$

and $H := y - m$ obeys $\mathrm{d}H = \lambda(\varepsilon - H)\,\mathrm{d}t$.

**Initial conditions.** $H(\tau) = 0$ is *forced*: the filter output is continuous and $m$ is defined as the value of $y$ at $\tau$. It is not a modelling choice. By contrast $\varepsilon(\tau)$ is free; the present model sets it to 0. We propose instead

$$\varepsilon(\tau) \sim \mathrm{N}\big(0,\ \sigma_\varepsilon^2/2\kappa\big) \quad \text{independent of } W_\varepsilon \tag{ss}$$

— the stationary law of (1). Call (ss) the *stationary start* and $\varepsilon(\tau) = 0$ the *cold start*.

**Why.** $\varepsilon$ represents fluctuations intrinsic to holding a value. Nothing makes such a process sit exactly at its mean at the instant maintenance begins, and no measurement constrains it. As Corollary 6 shows, the cold start is the sole source of a slow $(1/\kappa)$ transient in a system whose only other timescale is the fast filter $1/\lambda$; under (ss) the model's delay-period predictions become exact rather than asymptotic.

*Remark* 1. (ss) removes the mechanism that gives Model 1 its only delay-period ramp. That is not a way of disarming Model 1: Section 4 bounds what *any* residual non-stationarity can buy it, and the bound is far below the data.

**Notation.** $\rho := t - \tau$ is elapsed maintenance time; $\beta/\delta$ is the speed of temporal preparation, with $\beta = 0$ under Model 1; $d(t) = g - |m| - (\beta/\delta)(t - t_{\text{offset}})^+$ is the distance to the gate; $\varphi, \Phi$ are the standard normal density and c.d.f., and $g(u) := u\Phi(u) + \varphi(u)$, so $g' = \Phi$ and $g'' = \varphi$. Write $s^2 := \sigma_\varepsilon^2/2\kappa$ and

$$A_\infty := \frac{\lambda s^2}{\lambda + \kappa}, \qquad B_\infty := \frac{\lambda^2 \sigma_\varepsilon^2}{2(\lambda + \kappa)}, \qquad \omega := \sqrt{B_\infty/A_\infty} = \sqrt{\lambda\kappa}, \qquad a_\infty := \frac{d}{\sqrt{A_\infty}}. \tag{3}$$

## 2 Exact second-order moments

**Proposition 2** (exact moments)**.** *Under* (1)–(ss) *with* $H(\tau) = 0$*, put*

$$X(\rho) := \frac{(\lambda + \kappa)\big(e^{-2\lambda\rho} - e^{-(\lambda+\kappa)\rho}\big)}{\lambda - \kappa} \qquad \big(= -2\lambda\rho\, e^{-2\lambda\rho} \quad \textit{when } \lambda = \kappa\big). \tag{4}$$

*Then for every* $\rho > 0$*,*

$$\boxed{\frac{\sigma_H^2}{A_\infty} = 1 + X - e^{-(\lambda+\kappa)\rho}, \qquad \frac{c_H}{A_\infty} = -\lambda X, \qquad \frac{\sigma_{\dot H}^2}{B_\infty} = 1 + \frac{\lambda}{\kappa}\Big(X + e^{-(\lambda+\kappa)\rho}\Big).} \tag{5}$$

*In particular* $\sigma_H^2(0) = 0$*,* $c_H(0) = 0$*,* $\sigma_{\dot H}^2(0) = \frac{\lambda+\kappa}{\kappa} B_\infty$*, and all three converge to* (3) *with* $c_H(\infty) = 0$*.*

*Proof.* $(\varepsilon, H)$ is a two-dimensional linear diffusion with $F = \left(\begin{smallmatrix} -\kappa & 0 \\ \lambda & -\lambda \end{smallmatrix}\right)$ and $Q = \mathrm{diag}(\sigma_\varepsilon^2, 0)$, so its covariance solves $\dot P = FP + PF^\top + Q$ with $P(0) = \mathrm{diag}(s^2, 0)$. The system is triangular:

$$\dot P_{\varepsilon\varepsilon} = -2\kappa P_{\varepsilon\varepsilon} + \sigma_\varepsilon^2, \qquad P_{\varepsilon\varepsilon}(0) = s^2, \tag{6}$$

$$\dot P_{\varepsilon H} = -(\kappa + \lambda) P_{\varepsilon H} + \lambda P_{\varepsilon\varepsilon}, \qquad P_{\varepsilon H}(0) = 0, \tag{7}$$

$$\dot P_{HH} = 2\lambda(P_{\varepsilon H} - P_{HH}), \qquad P_{HH}(0) = 0. \tag{8}$$

By (6), $\dot P_{\varepsilon\varepsilon}(0) = -2\kappa s^2 + \sigma_\varepsilon^2 = 0$, so $P_{\varepsilon\varepsilon} \equiv s^2$: *the stationary start freezes the first component exactly.* Then (7) gives $P_{\varepsilon H} = A_\infty\big(1 - e^{-(\lambda+\kappa)\rho}\big)$, and (8), integrated from 0,

$$P_{HH} = 2\lambda \int_0^\rho e^{-2\lambda(\rho - v)} P_{\varepsilon H}(v)\mathrm{d}v = A_\infty \left[1 - e^{-2\lambda\rho} - 2\lambda \frac{e^{-(\lambda+\kappa)\rho} - e^{-2\lambda\rho}}{\lambda - \kappa}\right],$$

and writing $-2\lambda = -(\lambda+\kappa) - (\lambda-\kappa)$ in the numerator gives $P_{HH}/A_\infty - 1 = X - e^{-(\lambda+\kappa)\rho}$. Since $\dot{H} = \lambda(\varepsilon - H)$, $c_H = \lambda(P_{\varepsilon H} - P_{HH}) = -\lambda A_\infty X$ and $\sigma^2_{\dot{H}} = \lambda^2(P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}) = B_\infty + \lambda^2 A_\infty\big(X + e^{-(\lambda+\kappa)\rho}\big)$, using $\lambda^2(s^2 - A_\infty) = B_\infty$ and $\lambda^2 A_\infty/B_\infty = \lambda/\kappa$. The $\lambda = \kappa$ forms follow by l'Hôpital. □

*Remark* 3. $\sigma_H$ rises from 0; $\sigma_{\dot{H}}$ *falls* from $\frac{\lambda+\kappa}{\kappa}B_\infty$, because $\dot{H}(\tau) = \lambda\varepsilon(\tau)$ has the noise at full amplitude while $H$ is still at zero. The position variance fills in while the velocity variance settles down.

## 3 How fast the transient dies

**Lemma 4.** *For $A, B > 0$ and $\rho > 0$, $\left|e^{-A\rho} - e^{-B\rho}\right| \le |A - B|\,\rho\, e^{-\min(A,B)\rho}$.*

*Proof.* The mean value theorem applied to $x \mapsto e^{-x\rho}$ on the interval with endpoints $A, B$ gives $|A-B|\rho e^{-\xi\rho}$ for some $\xi$ between them, and $e^{-\xi\rho} \le e^{-\min(A,B)\rho}$. □

**Proposition 5** (transient bounds)**.** *Set $\mu := \lambda + \min(\lambda, \kappa)$, so $\lambda \le \mu \le \lambda + \kappa$, and*

$$\bar{\Delta}(\rho) := \big(1 + (\lambda+\kappa)\rho\big)e^{-\mu\rho}, \qquad \bar{\gamma}(\rho) := \lambda(\lambda+\kappa)\rho\, e^{-\mu\rho}. \tag{9}$$

*Then for every $\rho > 0$ and all $\lambda, \kappa > 0$: $0 \le -X(\rho) \le (\lambda+\kappa)\rho e^{-\mu\rho}$, and hence*

$$\left|\frac{\sigma^2_H}{A_\infty} - 1\right| \le \bar{\Delta}, \qquad 0 \le \frac{c_H}{A_\infty} \le \bar{\gamma}, \qquad \left|\frac{\sigma^2_{\dot{H}}}{B_\infty} - 1\right| \le \frac{\lambda}{\kappa}\bar{\Delta}. \tag{10}$$

*Moreover $\bar{\Delta}$ and $\bar{\gamma}$ are decreasing on $\rho \ge 1$ whenever $\lambda \ge 1$.*

*Proof.* Lemma 4 with $A = 2\lambda$, $B = \lambda + \kappa$ gives $|A - B| = |\lambda - \kappa|$ and $\min(A, B) = \mu$, so $|X| \le (\lambda+\kappa)\rho e^{-\mu\rho}$. The sign: if $\lambda > \kappa$ the numerator of (4) is negative and the denominator positive; if $\lambda < \kappa$ both change sign; so $X \le 0$. The three bounds follow from (5) and $e^{-(\lambda+\kappa)\rho} \le e^{-\mu\rho}$. For monotonicity, $\frac{\mathrm{d}}{\mathrm{d}\rho}\big[(1+c\rho)e^{-\mu\rho}\big] = e^{-\mu\rho}(c - \mu - \mu c\rho)$ with $c = \lambda + \kappa \ge \mu$; this is negative for $\rho > (c-\mu)/(\mu c)$, and $(c-\mu)/(\mu c) \le 1/(2\lambda) \le 1/2$ when $\lambda \ge 1$. Same for $\bar{\gamma}$. □

**Corollary 6** (relaxation is at least as fast as $\lambda$)**.** From Propositions 2 and 5. *All three moments converge to (3) at rate $\mu \ge \lambda$, uniformly in $\kappa$. The operator $P \mapsto FP + PF^\top$ has eigenvalues $\{-2\kappa, -(\lambda+\kappa), -2\lambda\}$, so a general initial condition relaxes as a combination of $e^{-2\kappa\rho}$ (the $\varepsilon$–$\varepsilon$ block), $e^{-(\lambda+\kappa)\rho}$ (the cross term) and $e^{-2\lambda\rho}$ (the $H$–$H$ block). Under (SS) the first has zero coefficient, because $P_{\varepsilon\varepsilon}$ is constant; the slower of the remaining two is $e^{-\mu\rho}$, which equals $e^{-2\lambda\rho}$ only when $\kappa \ge \lambda$.*

*Remark* 7 (the $\kappa$ floor). The first two bounds in (10) are uniform in $\kappa$; the third carries $\lambda/\kappa$ and is not. What it needs is weak: at $\rho = 1$ the deviation is $(\lambda/\kappa)e^{-2\lambda}$ to leading order, so relative accuracy $\epsilon$ needs $\kappa \gtrsim \lambda e^{-2\lambda}/\epsilon$, which at $\lambda = 10$, $\epsilon = 10^{-2}$ is $\kappa \ge 2 \times 10^{-6}$. Read through the correlation time: $\mathrm{Cov}(\varepsilon(t), \varepsilon(t+r)) \propto e^{-\kappa|r|}$ at stationarity, so $1/\kappa$ is the lag at which the maintenance noise decorrelates to $1/e$. A process with $1/\kappa$ of minutes is constant over the 3 s delay and over dozens of consecutive trials — a session-level drift, for which the model has no term. A floor $\kappa \ge 0.05\,\mathrm{s}^{-1}$ ($1/\kappa \le 20$ s, about seven delay periods) is therefore a statement about what the term means, not a range imported from simulation.

## 4 Neglecting the transient

Let $\nu_{\mathrm{stat}}$ denote the rate computed from the stationary moments (3), i.e. with $\sigma_H^2 = A_\infty$, $\sigma_H^2 = B_\infty$, $c_H = 0$. This section bounds the error committed by using $\nu_{\mathrm{stat}}$ in place of $\nu$.

**Theorem 8** (the cost of using the stationary rate)**.** *Fix $\rho \geq \rho_0 \geq 1$ and write $\bar{\Delta} := \bar{\Delta}(\rho_0)$, $\bar{\gamma} := \bar{\gamma}(\rho_0)$ of (9), $u_s := \beta/(\delta\sqrt{B_\infty})$, and*

$$\eta := \frac{\lambda}{\kappa}\bar{\Delta} + \frac{\bar{\gamma}^2}{\omega^2(1-\bar{\Delta})}. \tag{11}$$

*Assume $\lambda \geq 1$, $\bar{\Delta} < 1$ and $\eta < 1$. Then, for both models,*

$$\boxed{\left|\ln \nu(\rho) - \ln \nu_{\mathrm{stat}}(\rho)\right| \;\leq\; \mathcal{E} := \frac{(a_\infty^2+1)\bar{\Delta}}{2(1-\bar{\Delta})} + \frac{\eta}{2(1-\eta)} + \sqrt{2\pi}\left[\frac{\bar{\gamma}\, a_\infty}{\omega(1-\bar{\Delta})\sqrt{1-\eta}} + u_s\left(\frac{1}{\sqrt{1-\eta}} - 1\right)\right].} \tag{12}$$

*Proof.* Write $\Delta := \sigma_H^2/A_\infty - 1$, $\gamma := c_H/A_\infty$, $E := \sigma_H^2/B_\infty - 1$, so that by Proposition 5 and its monotonicity clause $|\Delta| \leq \bar{\Delta}$, $0 \leq \gamma \leq \bar{\gamma}$, $|E| \leq (\lambda/\kappa)\bar{\Delta}$ for all $\rho \geq \rho_0$. From the Kac–Rice formula (14) below,

$$\ln \nu - \ln \nu_{\mathrm{stat}} = -\tfrac{1}{2}\big(a^2 - a_\infty^2\big) - \tfrac{1}{2}\ln\frac{\sigma_H^2}{A_\infty} + \ln\frac{G}{G_s},$$

with $G = \sqrt{S}\, g(u)$, $G_s = \sqrt{B_\infty}\, g(u_s)$. We bound the three terms. Throughout, $|\ln(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$.

*(1)* $a^2 = d^2/\sigma_H^2 = a_\infty^2/(1+\Delta)$, so $\big|a^2 - a_\infty^2\big| = a_\infty^2|\Delta|/(1+\Delta) \leq a_\infty^2\bar{\Delta}/(1-\bar{\Delta})$, and the first term is at most $a_\infty^2\bar{\Delta}/\big(2(1-\bar{\Delta})\big)$.

*(2)* $\frac{1}{2}\big|\ln(1+\Delta)\big| \leq \bar{\Delta}/\big(2(1-\bar{\Delta})\big)$. Adding to (1) gives the first bracket of (12).

*(3a)* $S = \sigma_H^2 - c_H^2/\sigma_H^2 = B_\infty\big[(1+E) - \gamma^2/(\omega^2(1+\Delta))\big]$, using $A_\infty/B_\infty = \omega^{-2}$. Hence $|S/B_\infty - 1| \leq \eta$ with $\eta$ of (11), so $S \geq B_\infty(1-\eta) > 0$ and $\frac{1}{2}\big|\ln(S/B_\infty)\big| \leq \eta/\big(2(1-\eta)\big)$.

*(3b)* $M = \beta/\delta + \gamma d/(1+\Delta) \geq \beta/\delta \geq 0$, so $u = M/\sqrt{S} \geq 0$; also $u_s \geq 0$. On $[0,\infty)$, $g$ is increasing with $g \geq g(0) = 1/\sqrt{2\pi}$ and $g' = \Phi \leq 1$, so for $u, u_s \geq 0$

$$\big|\ln g(u) - \ln g(u_s)\big| \leq \frac{|u - u_s|}{\min_{[u\wedge u_s,\, u\vee u_s]} g} \leq \sqrt{2\pi}\,|u - u_s|.$$

Split $u - u_s = \dfrac{M - \beta/\delta}{\sqrt{S}} + \dfrac{\beta}{\delta}\Big(\dfrac{1}{\sqrt{S}} - \dfrac{1}{\sqrt{B_\infty}}\Big)$. For the first, $M - \beta/\delta = \gamma d/(1+\Delta) \leq \bar{\gamma} a_\infty\sqrt{A_\infty}/(1-\bar{\Delta})$ and $\sqrt{S} \geq \sqrt{B_\infty(1-\eta)}$, giving $\bar{\gamma}a_\infty/\big(\omega(1-\bar{\Delta})\sqrt{1-\eta}\big)$. For the second, it equals $u_s\big|(1+x)^{-1/2} - 1\big|$ with $x = S/B_\infty - 1$, $|x| \leq \eta$, hence at most $u_s\big(1/\sqrt{1-\eta} - 1\big)$. Summing gives (12). □

**Corollary 9** (what it costs on the measured quantities)**.** From Theorem 8. *Let $f$ be the log rate on the 20 bin centres of the window $[1,3]$ s, and $c_1, c_2$ the slope and quadratic coefficient of the least-squares quadratic fitted to it. These are linear functionals $c_j = \sum_i w_i^{(j)} f_i$ with $\sum_i |w_i^{(\mathrm{slope})}| = 1.504$ and $\sum_i |w_i^{(2)}| = 2.916$. Hence, if $|f - f_{\mathrm{stat}}| \leq \mathcal{E}$ pointwise,*

$$\big|c_1 - c_1^{\mathrm{stat}}\big| \leq 1.504\,\mathcal{E}\ \mathrm{s}^{-1}, \qquad \big|c_2 - c_2^{\mathrm{stat}}\big| \leq 2.916\,\mathcal{E}\ \mathrm{s}^{-2}. \tag{13}$$

*Remark* 10 (why this step is needed). Proposition 5 bounds the *moments*; the observables are a slope and a curvature, that is, *derivatives*, and a bound on a function does not bound its derivative. Theorem 8 bridges the gap by bounding $\ln \nu$ itself, pointwise, after which (13) is elementary because $c_1$ and $c_2$ are bounded linear functionals of the profile. Without this step, "the transient is negligible" is an assertion, not a theorem.

### 4.1 The conditions, and what justifies each

Theorem 8 needs exactly three things.

**(C1)** $\lambda$ **bounded below.** $1/\lambda$ is the time constant of the low-pass filter carrying motor preparation to the gate. External evidence: Dendauw et al. estimate $\lambda \approx 20\,\mathrm{s}^{-1}$ in an immediate-report task, and the present sweep halves it to $\lambda \geq 10$. It is in principle identifiable from these data, since $\lambda$ sets how fast the rate follows the stimulus: a filter with $1/\lambda > 0.7$ s would smear the stimulus-period coherence effect, which is observed sharply. *That check has not been carried out and would convert (C1) from a prior into a measurement.*

**(C2)** $\kappa$ **bounded below.** See Remark 7: $1/\kappa$ is the correlation time of the maintenance noise, and a value of minutes describes a session drift rather than a within-trial fluctuation. $\kappa \geq 0.05\,\mathrm{s}^{-1}$ is a statement about what the term means. (C1) and (C2) are substitutes: raising $\lambda$ by 2 buys roughly the same as raising the $\kappa$ floor by a factor of 4, so only one of the two needs to be argued hard.

**(C3)** $a_\infty$ **bounded above.** This is *measured*, not assumed — see Section 6: the observed rate determines $a$ through Model 1's own stationary formula, giving $a \leq 3.3$, and the slope and curvature give $a = 3.05$ independently.

With $\rho_0 = 1$ s, $a_\infty \leq 3.3$ and $\beta/\delta \leq 0.3$, $\mathcal{E}$ is:

| | $\kappa \geq 0.05$ | $\kappa \geq 0.22$ | $\kappa \geq 1$ |
|---|---|---|---|
| $\lambda$ | $(1/\kappa \leq 20$ s) | $(\leq 4.5$ s) | $(\leq 1$ s) |
| 8 | $9.5 \times 10^{-1}$ | $2.0 \times 10^{-1}$ | $4.0 \times 10^{-2}$ |
| 10 | $1.3 \times 10^{-1}$ | $3.6 \times 10^{-2}$ | $7.4 \times 10^{-3}$ |
| 12 | $2.2 \times 10^{-2}$ | $6.4 \times 10^{-3}$ | $1.3 \times 10^{-3}$ |
| 15 | $1.6 \times 10^{-3}$ | $4.5 \times 10^{-4}$ | $8.7 \times 10^{-5}$ |
| 20 | $1.7 \times 10^{-5}$ | $4.8 \times 10^{-6}$ | $9.0 \times 10^{-7}$ |

At $\lambda = 10$, $\kappa \geq 0.22$: $\mathcal{E} = 3.6 \times 10^{-2}$, so by (13) the stationary formula misstates the slope by at most $0.054\,\mathrm{s}^{-1}$ and the curvature by at most $0.105\,\mathrm{s}^{-2}$. *From here on we work with* $\nu_{\mathrm{stat}}$*, and carry* $1.504\,\mathcal{E}$ *as the error bar on every slope.*

## 5 The stationary rate

Let $Z := s_{\mathrm{resp}} H - d$, so that partial activations are upcrossings of zero by $Z$ and $\dot{Z} = \dot{H} + \beta/\delta$. The Kac–Rice formula for a differentiable Gaussian process gives

$$\nu(t) = p_{Z(t)}(0)\,\mathbb{E}\big[\dot{Z}\mathbf{1}_{\{\dot{Z}>0\}} \mid Z = 0\big] = \frac{\varphi(a)}{\sigma_H}\,G, \qquad G = M\Phi(u) + \sqrt{S}\varphi(u) = \sqrt{S}\,g(u), \quad (14)$$

with $a = d/\sigma_H$, $M = \beta/\delta + (c_H/\sigma_H^2)d$, $S = \sigma_{\dot{H}}^2 - c_H^2/\sigma_H^2$ and $u = M/\sqrt{S}$.

**Proposition 11** (the stationary rate)**.** *Substituting the stationary moments* (3) *(*$c_H = 0$*) into* (14)*,*

$$\nu_{\mathrm{stat}} = \varphi(a)\left[\frac{\beta}{\delta\sigma_H}\Phi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right) + \omega\,\varphi\left(\frac{\beta}{\delta\sigma_{\dot{H}}}\right)\right], \quad (15)$$

*whose bracket does not depend on* $d$*. Consequently:*

*(i)* ***Model 2*** *($\beta > 0$): $a(\rho) = a(0) - \beta\rho/(\delta\sigma_H)$ is affine, so exactly*

$$\frac{\mathrm{d}\ln\nu_{\mathrm{stat}}}{\mathrm{d}\rho} = \frac{\beta}{\delta\sigma_H}\,a(\rho), \qquad \frac{\mathrm{d}^2\ln\nu_{\mathrm{stat}}}{\mathrm{d}\rho^2} = -\left(\frac{\beta}{\delta\sigma_H}\right)^2, \tag{16}$$

*a downward parabola, the same one in all six conditions.*

*(ii)* ***Model 1*** *($\beta = 0$): $M = 0$, $u = 0$, $a$ constant, and*

$$\nu_{\mathrm{stat}} = \frac{\omega}{2\pi}\,e^{-a^2/2}, \qquad \frac{\mathrm{d}\ln\nu_{\mathrm{stat}}}{\mathrm{d}\rho} \equiv 0, \qquad \frac{\mathrm{d}^2\ln\nu_{\mathrm{stat}}}{\mathrm{d}\rho^2} \equiv 0. \tag{17}$$

*Proof.* Set $c_H = 0$ in (14); then $\ln\nu_{\mathrm{stat}} = \mathrm{const} - a^2/2$ and (16) follows from $a$ affine. For (ii) set $\beta = 0$: $M = 0$, $g(0) = 1/\sqrt{2\pi}$, $G = \sqrt{B_\infty}/\sqrt{2\pi} = \omega\sigma_H/\sqrt{2\pi}$, so $\nu_{\mathrm{stat}} = \varphi(a)\omega/\sqrt{2\pi} = \omega e^{-a^2/2}/2\pi$, constant in $\rho$. □

**Proposition 12** (level orders by distance)**.** From (15). *Under both models $\nu_{\mathrm{stat}} = \varphi(a) \times$ (a factor independent of $d$), so $\nu_{\mathrm{stat}}$ is strictly decreasing in $a$, with no condition whatever. Under Model 2 the slope $\beta a/(\delta\sigma_H)$ of (16) is strictly* increasing *in $a$. Hence level and slope order oppositely across conditions.*

*Remark* 13 (before the window). The same monotonicity holds during the transient, though it then needs a condition: from (14), $\partial\ln\nu/\partial d = -a/\sigma_H + c_H\Phi(u)/(\sigma_H^2 G)$ is negative iff $dG > c_H\Phi(u)$, and $G > M\Phi(u)$ with $dM = (\beta/\delta)d + c_H a^2 \geq c_H$ whenever $a \geq 1$. Not needed for anything below.

## 6 The standardised distance is measured

Inverting (17) — Model 1's own stationary rate formula, so using it to bound Model 1 is legitimate —

$$a = \sqrt{2\ln\Big(\frac{\omega}{2\pi\nu}\Big)}. \tag{18}$$

With the observed delay-period rates 0.030–0.135 $\mathrm{s}^{-1}$ and $(\lambda, \kappa)$ anywhere in the sampled range this gives $a \leq 3.3$, which is (C3). This is the rate–ramp trade-off in its sharpest form: the rate *measures* the standardised distance.

## 7 Predictions

The delay-period observations are: **(L1)** the level rises with duration; **(L2)** the level rises with coherence; **(L3)** the coherence effect on the level is larger after the long stimulus; **(S1)** the log-rate slope is $+0.867\,\mathrm{s}^{-1}$; **(S2)** it is shallower after the long stimulus, $0.896 \to 0.723$; **(S3)** shallower at high coherence, $0.871 \to 0.726$; **(C1)** the quadratic coefficient is $c_2 = -0.0405\,\mathrm{s}^{-2}$, not distinguishable from zero. Throughout, $\mathcal{E}$ is the error of Section 4.1 and the error bars are those of Corollary 9; at $\lambda = 10, \kappa \geq 0.22$ they are $0.054\,\mathrm{s}^{-1}$ on a slope and $0.105\,\mathrm{s}^{-2}$ on a curvature.

**Corollary 14** (L1, L2: the level)**.** From Proposition 12, plus the saturation argument of the main text, which concerns the distribution of $m$ and is untouched by (SS). *Duration and coherence each raise the fraction of trials whose maintained target has reached the cap $\theta$, hence lower $d$ in the sense of first-order stochastic dominance; since $\nu_{\mathrm{stat}}$ is decreasing in $d$ at every $\rho$, and averaging over trials preserves the ordering, both raise the level. Predicted by* both *models:* not discriminating.

**Corollary 15** (L3: the interaction)**.** From Proposition 12 and the same saturation argument. *As the capped fraction tends to one the coherence levels converge, and duration raises that fraction at every coherence, so the coherence effect on the level must* shrink *with duration. Predicted by both models, and* contradicted *by the data. Not discriminating; a genuine disagreement.*

**Corollary 16** (S1: the slope)**.** From Proposition 11 and Corollaries 9. *Model 1 predicts a slope of exactly* 0*, hence a measured slope of at most* $1.504\,\mathcal{E} = 0.054\,\mathrm{s}^{-1}$ *at* $\lambda = 10$*,* $\kappa \geq 0.22$ *(*0.19 *at* $\kappa \geq 0.05$*). Model 2 predicts* $\beta a/(\delta\sigma_H) > 0$*, its only source of a ramp.* Discriminating: the observed $+0.867\,\mathrm{s}^{-1}$ exceeds Model 1's ceiling by a factor of 4.6 to 16.

**Corollary 17** (S2, S3: the slope across conditions)**.** From Propositions 11 and 12, Corollaries 9 and 14. *Model 1 predicts a slope of zero in* every *condition, hence a condition difference of at most* $2 \times 1.504\,\mathcal{E} = 0.108\,\mathrm{s}^{-1}$ *at* $\lambda = 10, \kappa \geq 0.22$*. Model 2 predicts a difference of definite sign: the slope increases in a while the level decreases in a, so any manipulation that raises the level lowers the slope; by Corollary 14 duration and coherence both raise the level.* Discriminating: the observed differences 0.173 and 0.145 exceed Model 1's ceiling, and their sign is the one Model 2 requires.

**Corollary 18** (C1: the curvature)**.** From Proposition 11 and Corollary 9. *Model 1 predicts* $c_2 = 0$*, hence* $|c_2| \leq 2.916\,\mathcal{E} = 0.105\,\mathrm{s}^{-2}$*. Model 2 predicts*

$$c_2 = -\tfrac{1}{2}\left(\frac{\beta}{\delta\sigma_H}\right)^2 = -\tfrac{1}{2}\left(\frac{slope}{a}\right)^2 \;\geq\; -0.034 \tag{19}$$

*at the observed slope with* $a \leq 3.3$*, constant in* $\rho$ *and identical in all six conditions. The observed* $c_2 = -0.0405 \pm 0.0416$ *is consistent with both.* Not discriminating at the present precision. *Under the cold start the curvature* did *discriminate, through the decay that Model 1's variance filling forces; under* (ss) *that role passes to S2 and S3, and the curvature becomes a quantitative check on Model 2.*

**Corollary 19** (identification)**.** From Proposition 11(i) and (15); cross-checked against Section 6. *Under Model 2 the three delay-period observables determine three model quantities with no free parameter:*

$$\frac{\beta}{\delta\sigma_H} = \sqrt{-2c_2}, \qquad a = \frac{slope}{\sqrt{-2c_2}}, \qquad \omega \text{ from the level via (15).} \tag{20}$$

*With slope* $= 0.867$*,* $c_2 = -0.0405$*,* $\nu = 0.070$*:* $\beta/(\delta\sigma_H) = 0.285\,\mathrm{s}^{-1}$*,* $a = 3.05$ *and* $\omega = 45\,\mathrm{s}^{-1}$*. The value of a agrees with the independent estimate* $a \leq 3.3$ *from the level alone, a genuine check since the two routes use different observables. The value of* $\omega$ *is a prediction: it requires* $\lambda\kappa \approx 2.0 \times 10^3$*, above what the present sweep samples (*$\omega \leq 38$*). Caveat:* $c_2$ *is not significantly different from zero, so* (20) *is a point estimate whose interval on a, hence on* $\omega$*, is unbounded above.*

# 8 Summary

| | Model 1 | Model 2 | verdict |
|---|---|---|---|
| L1, L2 level | rises | rises | shared |
| L3 interaction | shrinks | shrinks | shared; both fail |
| S1 slope | 0 ($\leq 0.054$) | $\beta a/\delta\sigma_H > 0$ | **discriminating** |
| S2, S3 by condition | 0 ($\leq 0.108$) | sign forced | **discriminating** |
| C1 curvature | 0 ($\leq 0.105$) | $-\frac{1}{2}(\text{slope}/a)^2$ | not discriminating |

## 9 Numerical verification

| Claim | Test | Result |
|---|---|---|
| Prop. 2 | vs. numerical integration of $\dot{P} = FP + PF^\top + Q$ | max rel. err. $8.9 \times 10^{-12}$ |
| Prop. 2 | the three identities in terms of $X$ | agree to $1.5 \times 10^{-13}$ |
| Prop. 5 | $0 \le -X \le (\lambda+\kappa)\rho e^{-\mu\rho}$ (50-digit) | worst ratio 1.000000 |
| Prop. 5 | $\lvert\sigma_H^2/A_\infty - 1\rvert \le \bar{\Delta}$ | worst ratio 1.000000 |
| Prop. 5 | $c_H/A_\infty \le \bar{\gamma}$ | worst ratio 1.000000 |
| Prop. 5 | $\lvert\sigma_H^2/B_\infty - 1\rvert \le (\lambda/\kappa)\bar{\Delta}$ | worst ratio 0.997226 |
| Thm 8 | $\lvert\ln\nu - \ln\nu_{\text{stat}}\rvert \le \mathcal{E}$, 71 079 points | 0 violations, worst ratio 0.612 |
| Cor. 9 | the two functional norms 1.504, 2.916 | computed exactly |

Grids: $\lambda \in [0.3, 60]$, $\kappa \in [0.01, 60]$, $\rho \in [0.05, 6]$, $\beta/\delta \in \{0, 0.05, 0.3, 1, 3\}$, $a_\infty \in \{0.5, 1, 2, 3.3, 6, 10\}$. The $X$ bounds were verified in 50-digit arithmetic because the combination $P_{\varepsilon\varepsilon} - 2P_{\varepsilon H} + P_{HH}$ cancels catastrophically in double precision; points where $\mathcal{E}$ falls below double-precision resolution ($10^{-10}$) were excluded from the Theorem 8 check for the same reason.

## 10 Consequences for the paper

Adopting (ss) removes from the appendix the convolution kernel $K$, the integrals $\int K^2$ and $\int K'^2$, the auxiliary $W = K' - K^3/2\sigma_H^2$, the log-concavity lemma, the non-stationary derivative of $\ln\nu$, and the transient content of the present Corollaries A.2 and A.3. What remains is Sections 2–4 — five short results establishing that the transient may be neglected — followed by the stationary rate and the predictions as one-line corollaries.

The cost is that theory and simulation must agree, so the 1200-draw parameter sweep, the admissibility screen and every effect size would have to be recomputed under (ss).